

Lakshya JEE 2.0 2025

Mathematics

Indefinite Integration

DPP: 7

Q1 If $I_n = \int (\ln x)^n dx$, then $I_{10} + 10I_9$ is equal to
(where C is the constant of integration)

- (A) $x(\ln x)^{10} + C$
(B) $10(\ln x)^9 + C$
(C) $9(\ln x)^{10} + C$
(D) $x(\ln x)^9 + C$

Q2 Let $A_n = \int \tan^n x dx, \forall n \in N$. If $A_{10} + A_{12} = \frac{\tan^m x}{m} + \lambda$ (where λ is an arbitrary constant), then the value m is

- (A) 10 (B) 11
(C) 12 (D) 13

Q3 $\int \frac{dx}{(x+1)\sqrt{4x+3}}$ is equal to:

- (A) $\tan^{-1} \sqrt{4x+3} + c$
(B) $3 \tan^{-1} \sqrt{4x+3} + c$
(C) $2 \tan^{-1} \sqrt{4x+3} + c$
(D) $4 \tan^{-1} \sqrt{4x+3} + c$

Q4 $\int \frac{dx}{(1+x^2)\sqrt{1-x^2}}$

- (A) $\frac{-1}{\sqrt{4}} \tan^{-1} \left(\frac{\sqrt{1-x^2}}{\sqrt{2x}} \right) + c$
(B) $\frac{-1}{\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{1+x^2}}{\sqrt{2x}} \right) + c$
(C) $\frac{-1}{\sqrt{2}} \tan^{-1} \left(\frac{\sqrt{1-x^2}}{\sqrt{2x}} \right) + c$
(D) $\frac{1}{\sqrt{4}} \tan^{-1} \left(\frac{\sqrt{1-x^2}}{\sqrt{2x}} \right) + c$

Q5 $\int \frac{dx}{(x+1)\sqrt{x^2-1}}$

- (A) $\sqrt{\frac{x+1}{x-1}} + c$
(B) $-\sqrt{\frac{x-1}{x+1}} + c$
(C) $-\sqrt{\frac{x+1}{x-1}} + c$
(D) $\sqrt{\frac{x-1}{x+1}} + c$

Q6

$$\text{If } \int \frac{dx}{(x^2+x+1)^2} = a \cdot \tan^{-1} \left(\frac{2x+1}{\sqrt{3}} \right) + b \cdot \left(\frac{2x+1}{x^2+x+1} \right) + c, x > 0$$

then the value of $(\sqrt{3}a + b)$ is equal to:

Q7 Evaluate $\int (2x-4)\sqrt{4+3x-x^2} dx$

Q8 $\int \frac{x+2}{(x^2+3x+3)\sqrt{x+1}}$ is equal to

- (A) $\frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{x}{\sqrt{3(x+1)}} \right)$
(B) $\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{x}{\sqrt{3(x+1)}} \right)$
(C) $\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{x}{\sqrt{(x+1)}} \right)$
(D) None of these

Q9 $\int \frac{dx}{(x-1)^{3/4}(x+2)^{5/4}} =$

- (A) $\frac{2}{3} \left(\frac{x-1}{x+2} \right)^{1/4} + c$
(B) $\frac{1}{3} \left(\frac{x-1}{x+2} \right)^{1/4} + c$
(C) $\frac{4}{3} \left(\frac{x-1}{x+2} \right)^{1/4} + c$
(D) $\frac{2}{3} \left(\frac{x-1}{x-2} \right)^{1/4} + c$

Q10 $\int \frac{1-x^2}{1+x^2} \frac{dx}{\sqrt{1+x^2+x^4}}$

Q11 The value of $\int \frac{x^2-1}{(x^2+1)\sqrt{x^4+1}} dx$ equal to

- $\frac{1}{\sqrt{2}} \sec^{-1}(f(x)) + c$, then $f(x)$ is
(A) $\frac{x^2-1}{2x}$
(B) $\frac{x^2-1}{\sqrt{2x}}$
(C) $\frac{x^2-1}{2x}$
(D) $\frac{x^2+1}{\sqrt{2x}}$



- Q12** If $\int \frac{dx}{\sqrt[3]{\sin^{11} x \cos x}} = -\frac{a(b+c \tan^2 x)}{d \tan^2 x (\tan x)^{2/3}} + k$, then $a + b + c + d$ equal to
 (A) 8 (B) 16
 (C) 24 (D) 32

Q13
$$\int \frac{(1-x)dx}{(x+1)\sqrt{x^3+x^2+x}}$$

- (A) $\tan^{-1}\left(x + \frac{1}{x} + 1\right) + C$
 (B) $\tan^{-1} \sqrt{x + \frac{1}{x} + 1} + C$
 (C) $2 \tan^{-1} \sqrt{x + \frac{1}{x} + 1} + C$
 (D) None of these

Q14 If $\int \frac{(x+2)dx}{(x^2+3x+3)\sqrt{(x+1)}}$

$$= \frac{2}{\sqrt{3}} \left[\tan^{-1} \left(\frac{f(x)+1}{\sqrt{3}} \right) + \tan^{-1} \left(\frac{f(x)-1}{\sqrt{3}} \right) \right] + c$$

then $f(x)$ is equal to

- (A) $2\sqrt{x-1}$
 (B) $3\sqrt{x+1}$
 (C) $3\sqrt{x-1}$
 (D) $2\sqrt{x+1}$

Q15 If $\int (x^{2010} + x^{804} + x^{402})(2x^{1608} + 5x^{402} + 10)^{1/402} dx$
 $= \frac{1}{10a} (2x^{2010} + 5x^{804} + 10x^{402})^{\frac{a}{402}}$. Then $(a-400)$ is equal to

Q16 Let us consider the integral of the following forms $\int f(x_1, \sqrt{mx^2 + nx + p})$

Case I If $m > 0$, then put

$$\sqrt{mx^2 + nx + p} = u \pm x\sqrt{m}$$

Case II If $p > 0$, then put

$$\sqrt{mx^2 + nx + p} = ux \pm \sqrt{p}$$

Case III If quadratic equation

$$mx^2 + nx + p = 0$$

has real roots α and β there put

$$\sqrt{mx^2 + nx + p} = (x - \alpha) \text{ uor } (x - \beta)u$$

If $\int x^{-11} (1+x^4)^{-1/2} dx = \frac{1}{ax^{10}} (1+x^4)^m + \frac{1}{bx^6} (1+x^4)^n + \frac{1}{cx^2} (1+x^4)^p + k$
 where a, b, c, m, n, p are real number and k be arbitrary constant, then $\frac{a+b+c}{m+n+p}$ equal to

- (A) -2
 (B) $-\frac{1}{2}$
 (C) $\frac{1}{2}$
 (D) 2

Q17 The value of $\int \frac{dx}{\sqrt{(x-a)(b-x)}}$, is

- (A) $2 \sin^{-1} \sqrt{\frac{x-a}{b-a}} + C$
 (B) $2 \sin^{-1} \sqrt{\frac{x-b}{b-a}} + C$
 (C) $\sin^{-1} \sqrt{\frac{x-a}{b-a}} + C$
 (D) None of these

Q18 $\int \frac{\sqrt{1+x^2}}{1-x^2} dx = \left(\frac{\sqrt{2}}{2} \ln |f(x)| + g(x) + c \right)$, then

$f(x)$ and $g(x)$ are

- (A) $f(x) = \frac{\sqrt{1+x^2}-\sqrt{2}x}{\sqrt{1+x^2}+\sqrt{2}x}; g(x) = \ln(x + \sqrt{1+x^2}) + c$
 (B) $f(x) = \frac{\sqrt{1+x^2}+\sqrt{2}x}{\sqrt{1+x^2}-\sqrt{2}x}; g(x) = \ln(x - \sqrt{1+x^2}) + c$
 (C) $f(x) = \frac{\sqrt{1-x^2}-\sqrt{2}x}{\sqrt{1-x^2}+\sqrt{2}x}; g(x) = \ln(x + \sqrt{1-x^2}) + c$
 (D) $f(x) = \frac{\sqrt{1+x^2}+\sqrt{2}x}{\sqrt{1+x^2}-\sqrt{2}x}; g(x) = \ln(\sqrt{x^2-1} + x) + c$

Q19 $\int \left(\frac{\sin \theta}{\cos 3\theta} + \frac{\sin 3\theta}{\cos 9\theta} + \frac{\sin 9\theta}{\cos 27\theta} \right) d\theta$ is equal to

- (A) $\frac{1}{2} \log \left| \frac{\sec 27\theta}{\sec \theta} \right| + C$
 (B) $\frac{1}{2} \log \left| \frac{\sec \theta}{\sec 27\theta} \right| + C$
 (C) $\frac{1}{2} \log \left| \frac{\sec 27\theta}{\sec \theta} \right| + C$
 (D) None of these



Answer Key

Q1 (A)

Q2 (B)

Q3 (C)

Q4 (C)

Q5 (D)

Q6 $5/3$

Q7 $-\frac{2}{3}(4+3x-x^2)^{3/2}$
 $-\left\{\frac{2x-3}{4}\sqrt{4+3x-x^2} + \frac{25}{8}\sin^{-1}\frac{2x-3}{5}\right\}$
 $+c$

Q8 (B)

Q9 (C)

Q10 $-\tan^{-1}\sqrt{x^2+2+1/x^2-1}+c$

Q11 (D)

Q12 (A)

Q13 (D)

Q14 (D)

Q15 3

Q16 (A)

Q17 (A)

Q18 (B)

Q19 (C)



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